Planning Reliable Yellow-Taxi Journeys to JFK and LaGuardia

Estimating Travel-Time Allowances across Pickup Zones, Departure Times and Weather Conditions

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1 Introduction

For an airport passenger, underestimating travel time can mean a missed flight, while excessive caution creates unnecessary waiting. This report therefore asks: how much additional time should passengers allow for a yellow-taxi journey to John F. Kennedy International Airport (JFK) or LaGuardia Airport (LGA) under different pickup zones, departure times and weather conditions? The target audience is passengers deciding when to leave for the airport, for whom reliable upper-tail travel times are more informative than a single average.

This analysis uses six months of New York City Taxi and Limousine Commission trip records, from January to June 2024[1]. This recent six-month period provides a sufficiently large dataset while capturing winter-to-early-summer weather and daily travel patterns. Yellow taxis were selected because their records provide consistent pickup and drop-off timestamps and Taxi Zone identifiers required for spatial travel-time analysis. Trips ending in Taxi Zones 132 and 138 are assumed to represent journeys to JFK and LaGuardia, respectively. Temporal features, pickup zones, destination airport and weather variables were chosen because they describe conditions that passengers can identify before departure. NOAA observations from the JFK and LGA airport stations provide the external weather data [2]; as a simplifying assumption, each trip is matched by departure hour to the weather station as its destination airport. The resulting allowance concerns taxi travel only and excludes check-in, security and walking time within the airport.

The full trip distribution is first examined to identify missing, implausible and extreme records, leaving 400,942 trips for analysis. Descriptive and geospatial analyses then compare travel-time patterns across pickup zones, departure periods, destination airports and weather conditions. A Gamma generalised linear regression (Gamma GLR) and Gradient-Boosted Trees (GBT) are evaluated using expanding-window temporal windows, with June retained as the final future test period. Model predictions are then translated into practical travel-time allowances using the upper tail of earlier out-of-sample prediction errors.

2 Preprocessing

2.1 Taxi Data

- Six monthly Yellow Taxi Parquet files were obtained from the New York City Taxi and Limousine Commission (TLC) [1] and combined, producing 20,332,093 rows and 20 variables. Restricting pickup timestamps to January–June 2024 removed 39 out-of-period records, leaving $20{,}332{,}054 \times 20$.
- Trips ending in Taxi Zones 132 or 138 were selected. The official TLC Taxi Zone Lookup identifies these zones as JFK Airport and LaGuardia Airport respectively [1]. The airport filter produced $462{,}474 \times 20$ records, while joining the airport names produced $462{,}474 \times 21$: 227,833 trips to JFK and 234,641 to LaGuardia.
- No missing values occurred in the timestamps, zone identifiers or distance field required for the analysis. Five unused variables (passenger count, rate code, store-and-forward flag, congestion surcharge and airport fee) each contained 37,690 missing values (8.15%); these records were retained because the affected variables were not required for the research goal.
- Trip duration was calculated from the pickup and drop-off timestamps. Durations outside 1–180 minutes and distances above 100 miles were removed as implausible records. Trips with identical pickup and destination zones were excluded because they did not represent inbound journeys from another Taxi Zone. Zero-distance trips were retained when their duration and zone information remained plausible. No exact duplicates were found. These rules removed 60,496 records (13.08%), leaving $401{,}978 \times 22$.
- Calendar features were derived from pickup time, and pickup borough, zone and service-zone names were added from the TLC lookup. This produced $401{,}978 \times 32$ cleaned taxi records.

2.2 Weather Data

- Hourly weather observations were obtained from NOAA's Local Climatological Data [2]. The raw files contained $13{,}048 \times 125$ observations for JFK and $13{,}418 \times 125$ for LaGuardia. Retaining eight relevant fields, combining the stations and restricting observations to the study period produced $13{,}397 \times 8$.
- Observations were assigned to the nearest clock hour. Multiple observations within an hour were aggregated using mean temperature, wind speed and relative humidity, maximum precipitation and minimum visibility. Weather-event indicators were derived from precipitation, visibility and present-weather codes, producing $8{,}714 \times 14$ airport-hour records.
- Trace precipitation was coded as zero. Missing precipitation amounts were retained through a missing indicator and set to zero only in a separate modelling variable, avoiding the removal of otherwise complete weather hours.
- A left join by destination airport and pickup hour preserved all 401,978 cleaned trips and produced $401{,}978 \times 44$. Weather was matched to 401,239 trips (99.82%). Removing unmatched trips and records with missing temperature, visibility, wind speed or relative humidity left $400{,}942 \times 44$; 1,036 trips (0.26%) were removed at this stage.
- Finally, 28 variables relevant to the research goal were retained, yielding the $400{,}942 \times 28$ analysis dataset used in subsequent sections.

3 Analysis and Geospatial Visualisation

3.1 Travel-Time Distribution

Travel times differed substantially between the two airports. The median trip to JFK took 50.15 minutes, compared with 28.35 minutes for LaGuardia. The corresponding 95th percentiles were 84.88

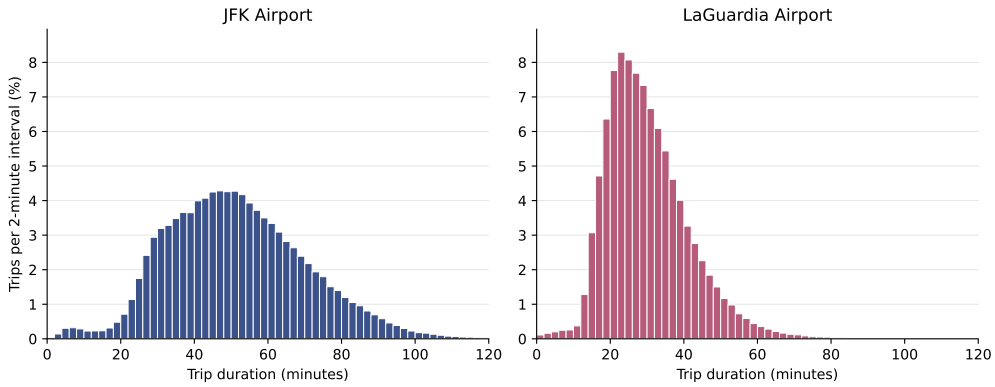


Figure 1: Distribution of taxi travel times to JFK and LaGuardia airports.

and 51.00 minutes, indicating historical travel-time buffers of 34.73 minutes for JFK and 22.65 minutes for LaGuardia when measured as the difference between the 95th percentile and the median.

Figure 1 shows that both distributions were right-skewed, with a longer upper tail for JFK. Therefore, relying only on the median would understate the time required during unusually slow journeys, particularly for passengers travelling to JFK.

3.2 Departure-Time Patterns

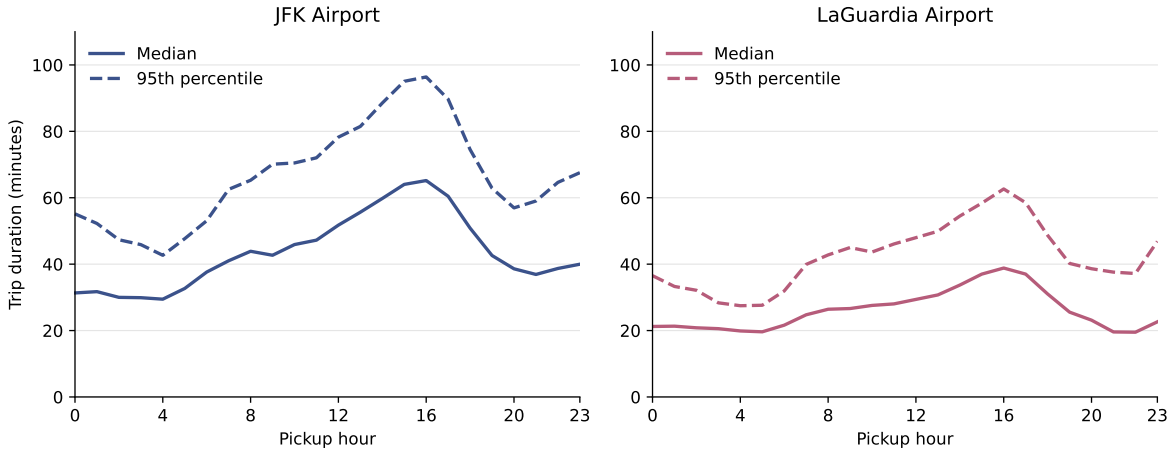


Figure 2: Median and 95th-percentile travel times by pickup hour and destination airport.

Travel times varied considerably across the day. For both airports, median and 95th-percentile travel times generally increased from the morning and peaked during the afternoon. At 16:00, JFK recorded a median of 65.18 minutes and a 95th percentile of 96.38 minutes, while the corresponding values for LaGuardia were 38.83 and 62.65 minutes.

The gap between the median and 95th percentile also widened during the busier daytime periods, indicating greater travel-time uncertainty. Departure time therefore affects not only the expected journey duration but also the additional time passengers should allow for potential delays.

3.3 Weather Conditions

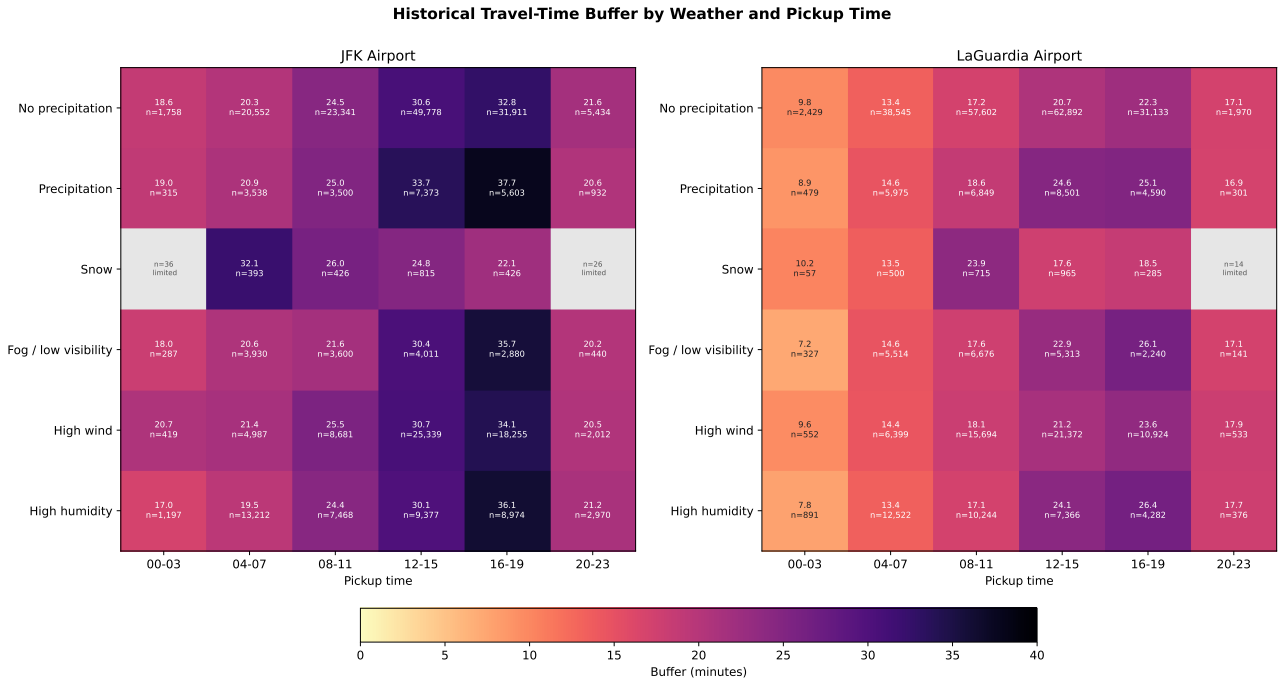


Figure 3: Historical travel-time buffer by weather condition, pickup period and destination airport. Each buffer is calculated as the 95th percentile minus the median; cells also report the number of trips.

Figure 3 shows that travel-time buffers were generally largest between 12:00 and 19:00, particularly for JFK. During 16:00–19:00 the JFK buffer increased from 32.8 minutes without precipitation to 37.7 minutes with precipitation. For LaGuardia, precipitation increased the buffer from 20.7 to 24.6 minutes during 12:00–15:00 and from 22.3 to 25.1 minutes during 16:00–19:00. Low visibility, high wind and high humidity were also associated with relatively large afternoon buffers.

These comparisons are descriptive because weather categories can overlap; for example, a trip may occur during both precipitation and high humidity. Some snow-period cells also contain relatively few trips and should therefore be interpreted cautiously. Nevertheless, the results suggest that adverse weather can amplify travel-time uncertainty, especially during already congested periods.

3.4 Pickup-Zone Differences

Figure 4 reveals substantial spatial variation in travel-time uncertainty. For trips to JFK, the largest buffers among eligible zones were concentrated in Lower Manhattan, including Greenwich Village North (36.5 minutes), the Lower East Side (36.0 minutes) and the Financial District North (35.4 minutes). By contrast, South Ozone Park, located closer to JFK, had a buffer of only 8.1 minutes.

A similar spatial pattern appeared for LaGuardia. The World Trade Center and TriBeCa/Civic Center had buffers of 30.1 and 29.1 minutes, whereas several northern Manhattan zones had buffers below 15 minutes. These differences indicate that pickup location affects more than the typical journey length: longer and more congestion-sensitive routes also require larger allowances for uncertainty. To reduce instability from small samples, only zones with at least 1,000 trips were coloured on the map.

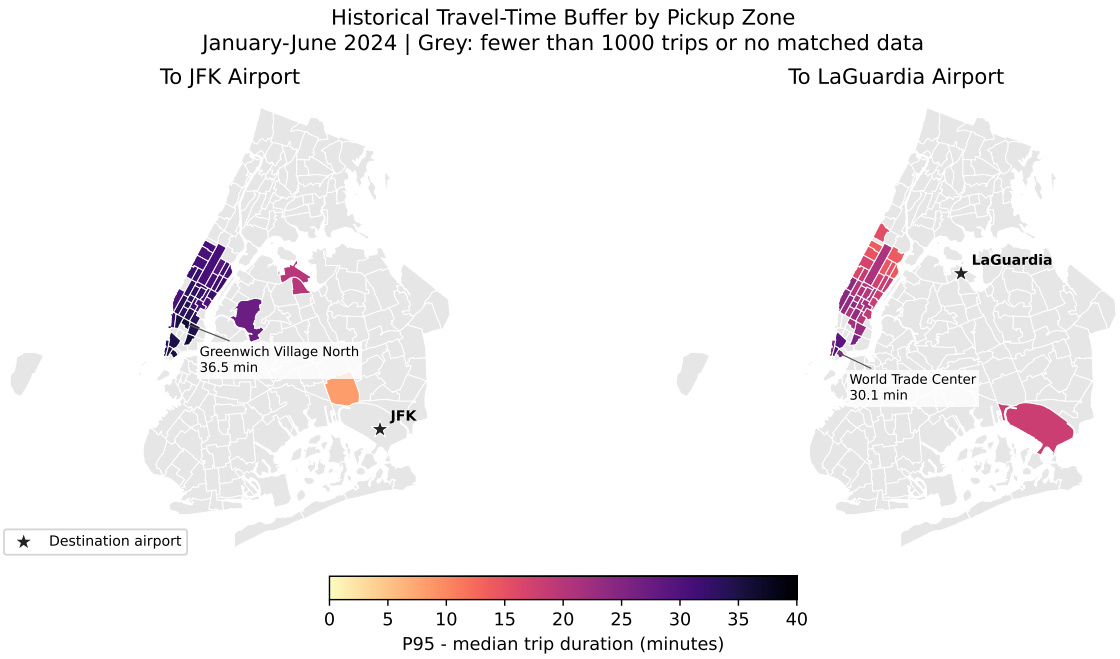


Figure 4: Historical travel-time buffer by pickup zone and destination airport. Grey zones contain fewer than 1,000 trips or have no matched data.

The exploratory analysis identifies destination airport, pickup time and pickup zone as important sources of travel-time variation. JFK journeys were generally longer and required larger buffers than LaGuardia journeys, while afternoon departures and adverse weather were associated with greater uncertainty. The next section evaluates whether these patterns remain useful when the predictors are considered jointly in statistical models.

4 Modelling

4.1 Model Design and Validation

Trip duration was modelled using destination airport, pickup zones, pickup hour, day of week and the available weather variables. Categorical predictors were one-hot encoded, while the weather predictors included temperature, precipitation, visibility, wind speed, relative humidity and indicators for precipitation, snow and low visibility. Actual trip distance was excluded because it is unknown before departure.

Four candidate models were initially evaluated: ridge regression[3], a Gamma generalised linear model (Gamma GLR)[4], random forest[5] and gradient-boosted trees (GBT)[6]. Gamma GLR was included because trip duration is positive and right-skewed, while GBT can represent nonlinear relationships and interactions without requiring them to be specified in advance. Based on the April evaluation, Gamma GLR and GBT were kept as the strongest parametric and tree-based models.

An expanding-window temporal validation design was used to preserve the order of the data and evaluate performance on future trips[7]. Models trained on January–March were evaluated on April, after which models trained on January–April were evaluated on May. The final models were trained on January–May and tested once on the held-out June data. The respective evaluation sets contained 67,929, 80,310 and 74,926 trips. This design assessed performance on future journeys and avoided

using later observations to predict earlier ones.

4.2 Model Performance

Table 1: Model performance in the rolling validation periods.

Evaluation period	Model	MAE	RMSE	R^2	Mean underprediction
April validation	Gamma GLR	6.772	9.414	0.724	-0.510
April validation	Ridge Regression	7.473	10.072	0.684	-1.643
April validation	Gradient-Boosted Trees	7.586	10.572	0.651	0.098
April validation	Random Forest	8.756	11.976	0.553	0.507
May validation	Gamma GLR	8.120	11.776	0.676	3.505
May validation	Gradient-Boosted Trees	9.108	12.870	0.613	3.870
June test	Gamma GLR	7.049	9.832	0.732	-0.616
June test	Gradient-Boosted Trees	8.089	11.164	0.654	0.067

Table 1 shows that Gamma GLR achieved the strongest performance in both validation periods. In April, it produced an MAE of 6.77 minutes, an $RMSE$ of 9.41 minutes and an R^2 of 0.724. Its performance weakened in May, when the MAE increased to 8.12 minutes and R^2 declined to 0.676, but it continued to outperform GBT.

On the held-out June test set, Gamma GLR achieved an MAE of 7.05 minutes, an $RMSE$ of 9.83 minutes and an R^2 of 0.732. GBT achieved an MAE of 8.09 minutes, an $RMSE$ of 11.16 minutes and an R^2 of 0.654. Gamma GLR therefore provided the more accurate and temporally consistent predictions, despite GBT's greater flexibility.

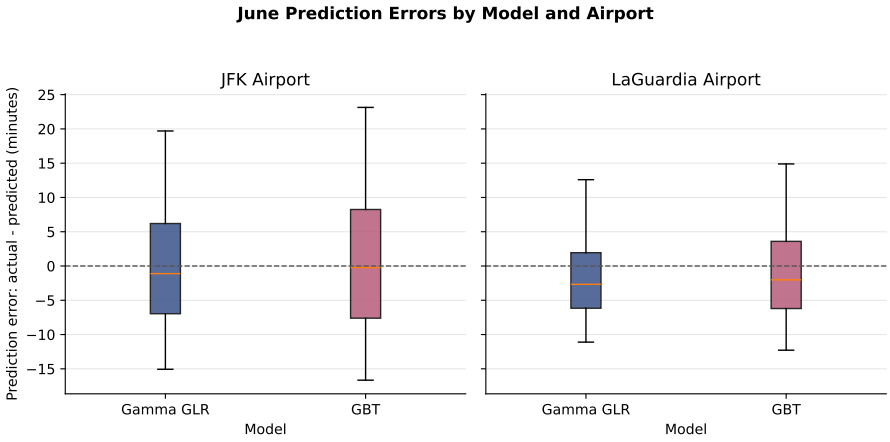


Figure 5: Distribution of June prediction errors by model and destination airport. Positive errors indicate underprediction. The boxes contain 50% of errors, while the lines extend from the 5th to 95th percentile.

Figure 5 shows wider prediction-error distributions for JFK than for LaGuardia, reflecting the greater variability of JFK journeys. Both models occasionally underpredicted long journeys, although Gamma GLR produced a narrower error distribution than GBT. Its mean June error was -0.62 minutes, indicating only slight overall overprediction, while the GBT mean error was close to zero.

4.3 Contribution of Weather Information

To assess weather information improved prediction, each selected model was refitted without the weather predictors while retaining destination airport, pickup zone, pickup hour and day of week. This comparison was conducted on the May validation period so that the June test set remained reserved for final evaluation.

Removing weather information reduced the performance of both models, although the size of the change differed. For Gamma GLR, the MAE increased from 8.12 to 8.39 minutes, the $RMSE$ increased from 11.78 to 12.25 minutes, and R^2 declined from 0.676 to 0.649. Weather information also reduced its mean underprediction from 4.84 to 3.51 minutes. For GBT, the improvements were smaller: MAE decreased from 9.14 to 9.11 minutes and R^2 increased from 0.605 to 0.613 when weather predictors were included.

Weather therefore provided additional predictive information, particularly for Gamma GLR, but its contribution was modest compared with the combined influence of pickup location and departure time. This is consistent with the exploratory analysis, where the largest weather-related differences occurred during particular time periods rather than uniformly throughout the day.

4.4 Travel-Time Allowance

Table 2: Travel-time allowances derived from earlier validation errors.

Model	Buffer (minutes)	Mean prediction	Mean recommended time	June coverage (%)
Gamma GLR	20.68	41.41	62.08	96.99
Gradient-Boosted Trees	22.47	40.72	63.19	96.54

To convert the predictions into practical travel-time allowances, the 95th percentile of the underprediction errors from the April and May validation periods was added to each model's June prediction:

$$\text{recommended taxi time} = \text{predicted taxi time} + \text{error buffer}$$

The Gamma GLR required a buffer of 20.68 minutes. When this buffer was added to its individual predictions, the mean recommended taxi time was 62.08 minutes and 96.99% of June journeys were covered. GBT required a larger buffer of 22.47 minutes and produced a mean recommendation of 63.19 minutes, with slightly lower June coverage of 96.54%.

These buffers represent additional time added to each personalised model prediction, rather than the complete taxi journey time. Gamma GLR therefore provided a slightly shorter average recommendation while covering a greater proportion of June journeys.

Overall, Gamma GLR provided useful but imperfect predictions. It explained 73.2% of the variation in June travel times with an mean absolute error of 7.05 minutes, compared with an overall average journey of 39.74 minutes. Since errors of this size remain important for airport travel, the prediction should be combined with an additional reliability buffer.

5 Recommendations

1. **Add approximately 21 minutes to the personalised travel-time prediction.** Passengers should plan their taxi journey using the Gamma GLR prediction plus a 20.68-minute reliability buffer, rather than using the prediction or a city-wide average alone, while the buffered recommendation covered 96.99% of June journeys (Table 2). The additional 21 minutes reduces

the risk that an unusually slow journey is underestimated without requiring every passenger to use the same total journey time.

2. **Update the estimate before afternoon trips to JFK, especially in rain or snow.** Passengers should use the latest available weather information and recalculate their departure time shortly before leaving. At 16:00, the median journey to JFK was 65.18 minutes but the 95th percentile reached 96.38 minutes (Figure 2). During 16:00–19:00, wet weather increased the historical JFK buffer from 32.8 to 37.7 minutes (Figure 3), while including weather information reduced the Gamma GLR's May underprediction from 4.84 to 3.51 minutes. If a personalised estimate is unavailable, a passenger travelling to JFK around 16:00 should therefore allow approximately 95–100 minutes for the taxi journey alone.

3. **Use the exact pickup zone instead of a general borough-level estimate.** Passengers should base their allowance on their specific Taxi Zone, because nearby origins can have substantially different delay risks. For journeys to JFK, the historical buffer was 36.5 minutes in Greenwich Village North, 36.0 minutes in the Lower East Side and 35.4 minutes in Financial District North, compared with only 8.1 minutes in South Ozone Park (Figure 4). Consequently, a single "Manhattan to JFK" allowance would be too general and could materially underestimate the time required from some pickup locations.

6 Limitations

This study covers Yellow Taxi journeys from January to June 2024, so the results may not generalise to other services, years or seasons. Airport weather may not represent conditions along the complete route, and overlapping weather categories prevent causal interpretation. The models also exclude real-time traffic, collisions, road closures and construction, which may produce unexpected delays. The 96.99% June coverage reflects historical performance rather than a guarantee, and the recommendations cover only the taxi journey; passengers must separately allow time for waiting and movement within the airport.

7 Conclusion

A single fixed travel-time estimate is not suitable for all airport journeys. Gamma GLR provided the most accurate and consistent predictions, and adding a 20.68-minute buffer to each trip-specific prediction covered 96.99% of journeys in the held-out June data. Passengers should therefore use an estimate based on their destination airport, pickup zone, departure time and expected weather, followed by an additional allowance of approximately 21 minutes. More conservative planning remains necessary for afternoon trips to JFK, rain or snow, and pickup zones with greater travel-time uncertainty.

References

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